

Final Report for STAT 451/551

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Introduction

The given FP data contains 200 observations of eight predictors labeled x_1 through x_8 , and the resulting response parameter y . Additional consideration was given to predicting y_{bin} which is 1 if y is greater than or equal to 70 and 0 otherwise. x_1 through x_7 are continuous while x_8 is categorical with four groups A, B, C , and D . No context surrounding the predictors or response parameters was given. This report is segmented into initial Exploratory Data Analysis, followed by the Regression Models, then the Classification Models, and concludes with some overall observations. All written code can be found in the appendix.

Exploratory Data Analysis

Continuous y

Correlations

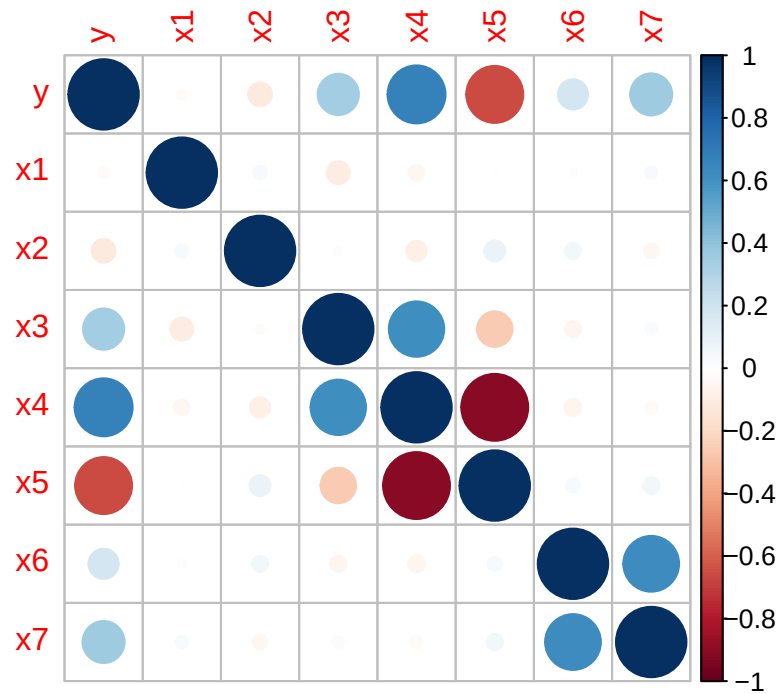


Figure 1: Correlations of Continuous Parameters

Pairplot

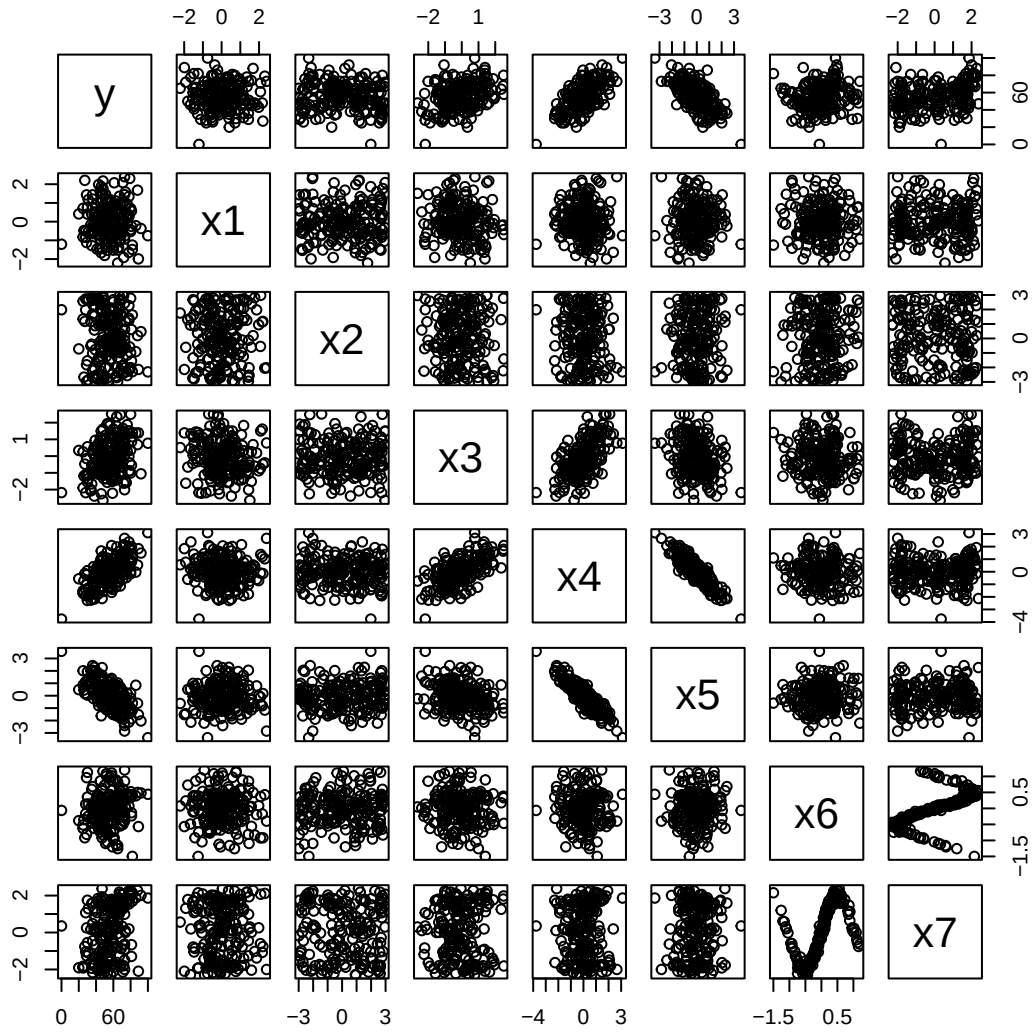


Figure 2: Scatterplots of Continuous Parameters

y

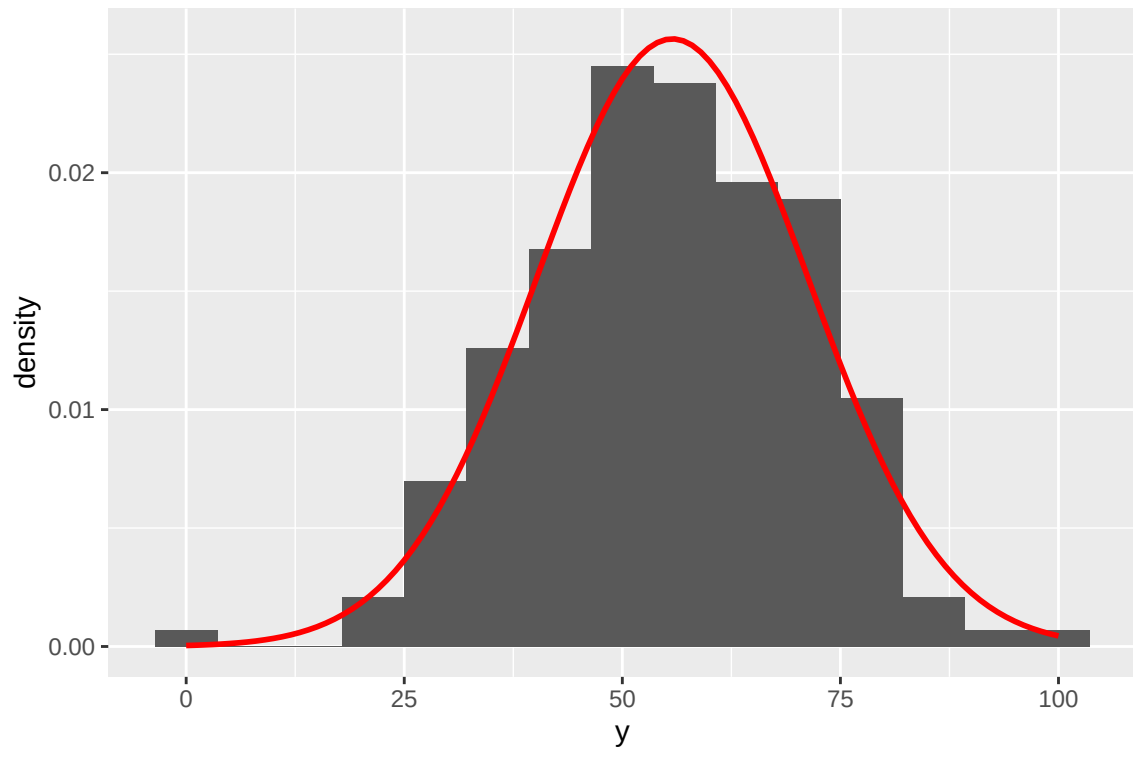


Figure 3: Histogram of y

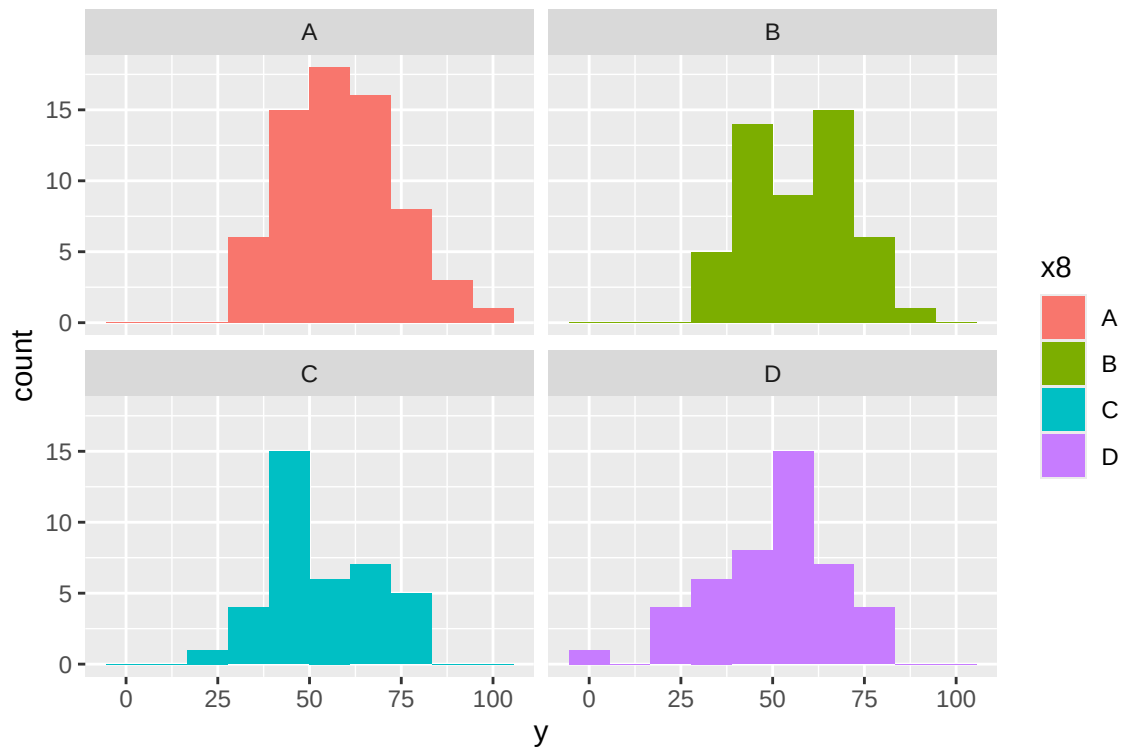


Figure 4: Histograms of y by x_8

- A right skewed
- D left skewed

Dichotomized y

Regression Models

Plotting the 10-Fold Cross Validation Metrics

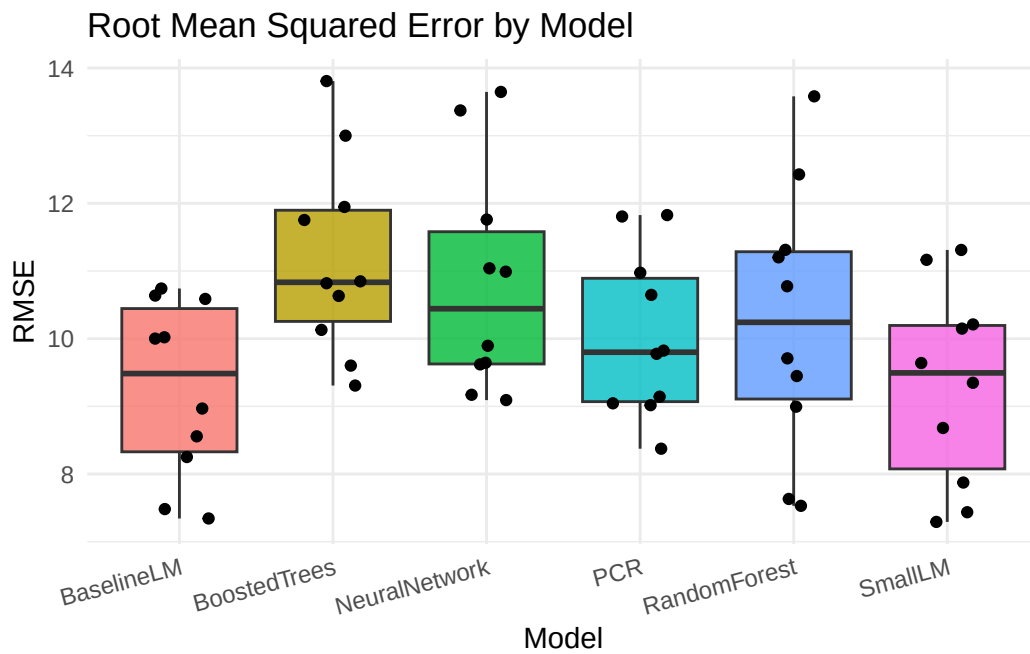


Figure 1: RMSE by Model for Each Fold

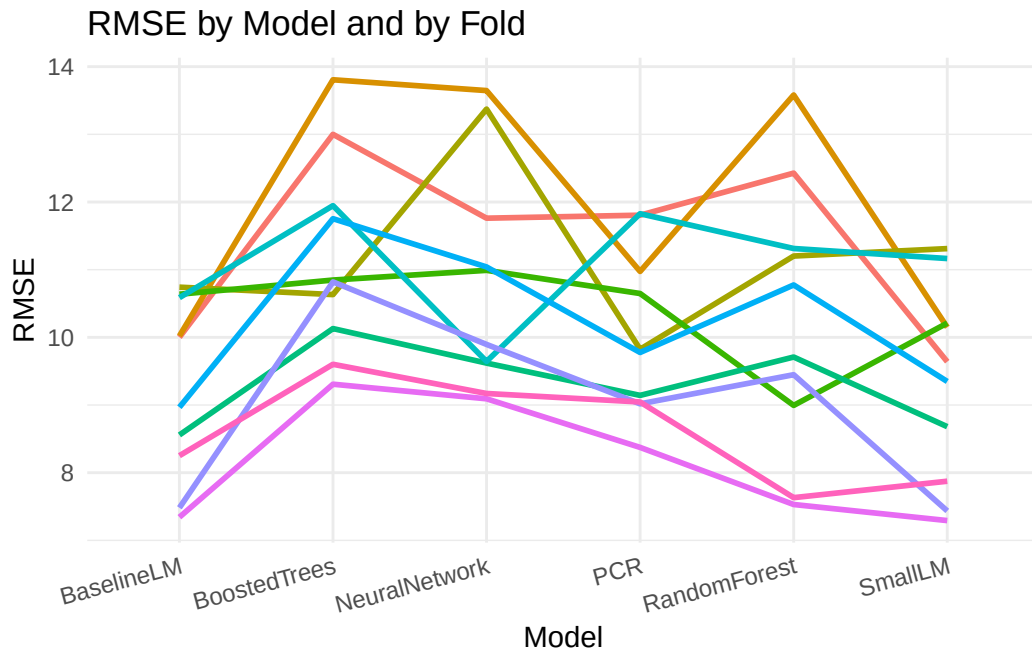


Figure 2: RMSE by Model and by Fold

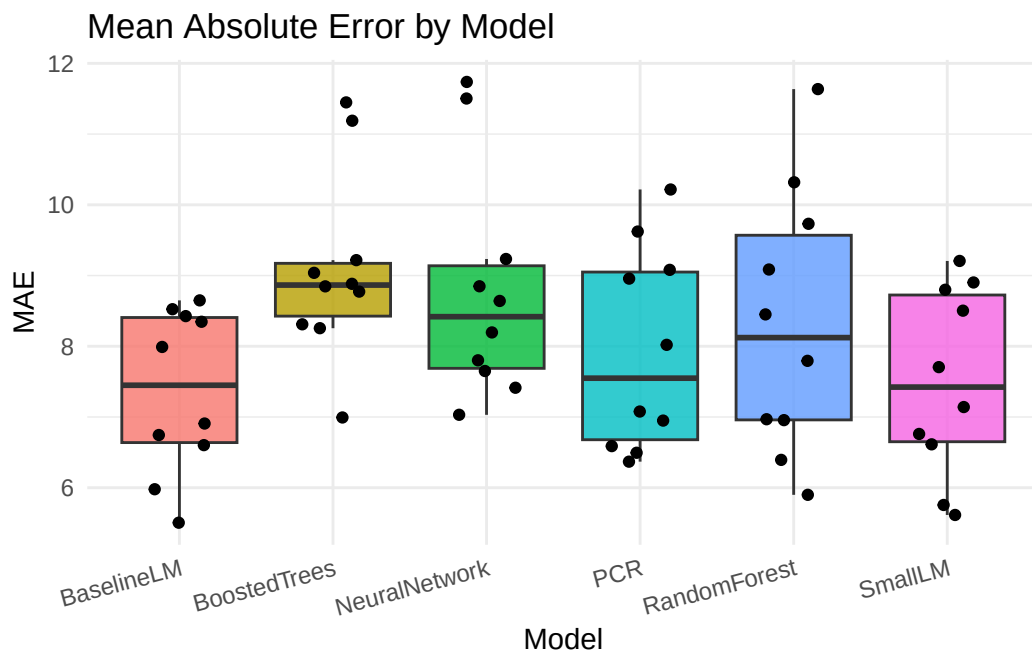


Figure 3: MAE by Model for Each Fold

Classification Models

Plotting the 10-Fold Cross Validation Metrics

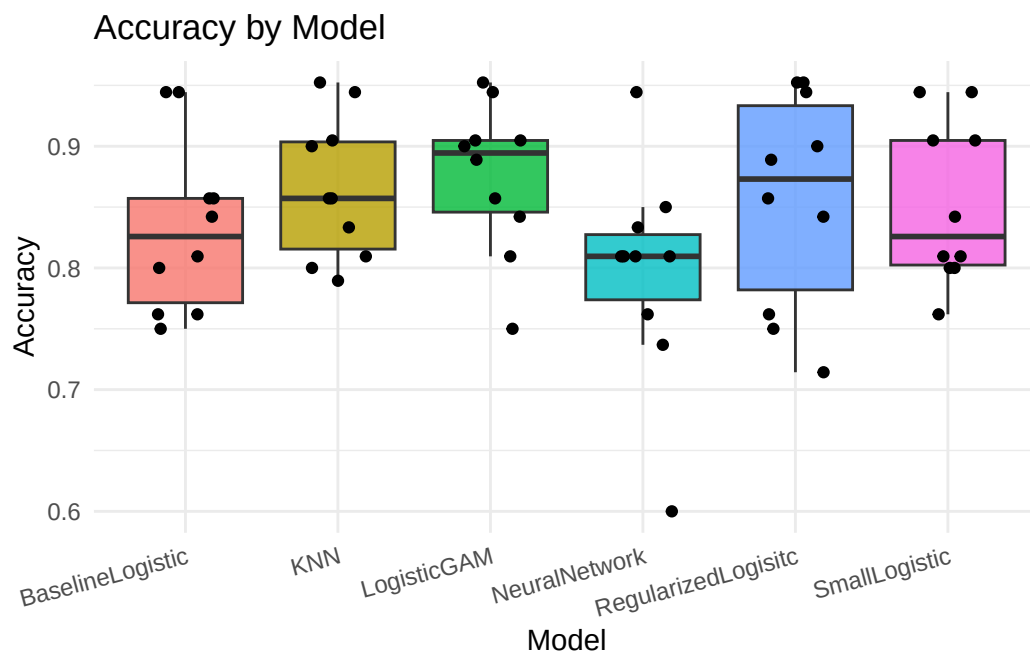


Figure 1: Accuracy by Model for Each Fold

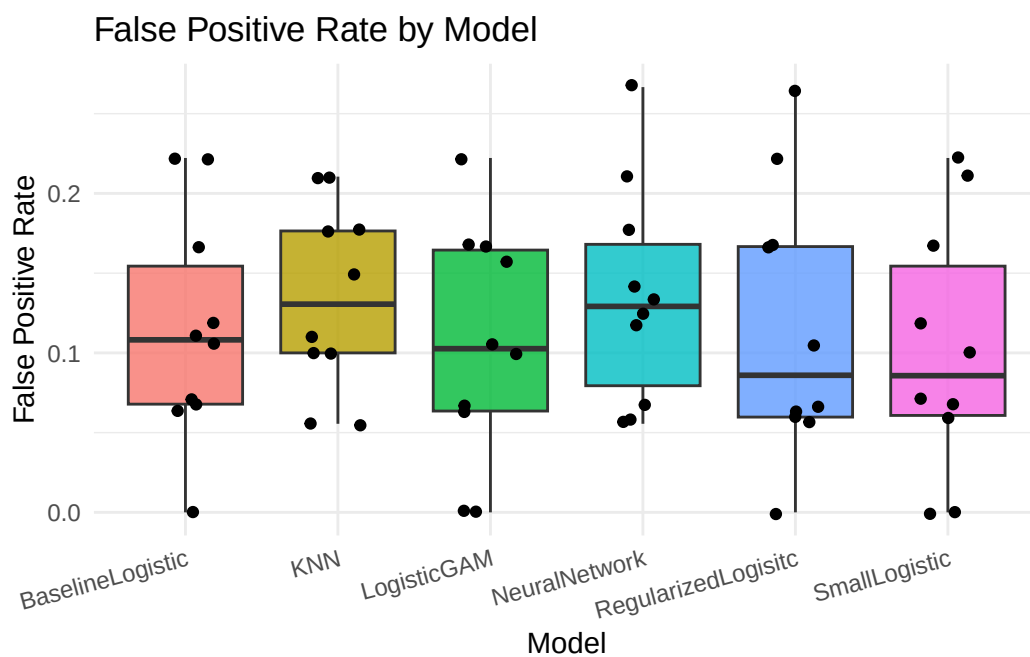


Figure 2: False Positive Rate by Model for Each Fold

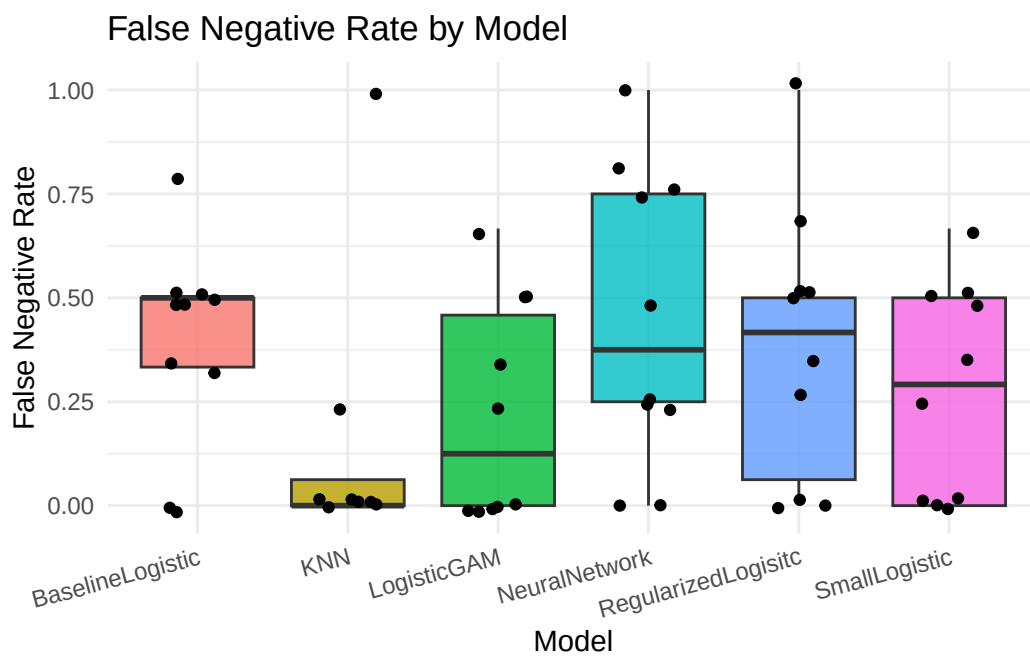


Figure 3: False Negative Rate by Model for Each Fold

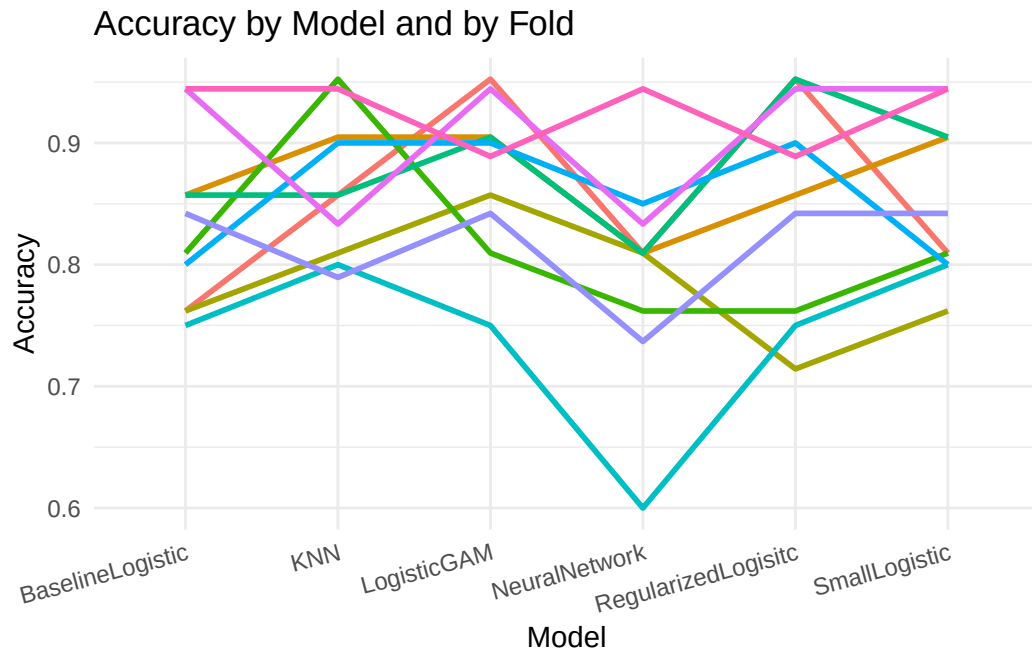


Figure 4: Accuracy by Model and by Fold